

Quasi-Interior Points in the Ordered Sobolev Spaces $W^{k,p}(\mathbb{R}^d)$

Automatic math-discovery packet

June 20, 2026

Source Question

The source paper is Jochen Glück and Martin R. Weber, *Almost Interior Points in Ordered Banach Spaces and the Long-Term Behaviour of Strongly Positive Operator Semigroups*, arXiv:1901.03306. In Example 2.20, after defining the ordered Sobolev space

$$W^{k,p}(\mathbb{R}^d)_+ = \{f \in W^{k,p}(\mathbb{R}^d) : f(\omega) \geq 0 \text{ for a.e. } \omega \in \mathbb{R}^d\},$$

the authors write:

We do not know whether the space $W^{k,p}(\mathbb{R}^d)$ has almost interior or even quasi-interior points.

Here $k \in \mathbb{N}_0$ and $1 \leq p < \infty$. In the terminology recalled in the source paper, a positive vector u is a quasi-interior point if the principal ideal

$$X_u = \{y \in X : \exists \lambda \geq 0 \text{ such that } -\lambda u \leq y \leq \lambda u\}$$

is dense in X . Every quasi-interior point is almost interior.

Idea of the Proof

The order is pointwise almost everywhere, so a strictly positive smooth function can dominate compactly supported smooth functions after multiplying by a scalar. The Gaussian $u(x) = e^{-|x|^2}$ is the natural choice: it belongs to every finite-order Sobolev space $W^{k,p}(\mathbb{R}^d)$, it is positive everywhere, and on each compact set it has a positive lower bound. Since $C_c^\infty(\mathbb{R}^d)$ is dense in $W^{k,p}(\mathbb{R}^d)$, the principal ideal generated by u is dense.

The Result

Theorem 1. *Let $d \geq 1$, $k \in \mathbb{N}_0$, and $1 \leq p < \infty$. Endow $W^{k,p}(\mathbb{R}^d)$ with the cone of almost everywhere nonnegative functions. Then*

$$u(x) = e^{-|x|^2}$$

is a quasi-interior point of $W^{k,p}(\mathbb{R}^d)_+$. Consequently $W^{k,p}(\mathbb{R}^d)_+$ has almost interior points.

Proof. First, $u \in W^{k,p}(\mathbb{R}^d)$. Indeed, every weak derivative of u of order at most k is a polynomial times $e^{-|x|^2}$, hence belongs to $L^p(\mathbb{R}^d)$ for every finite p .

We show that $C_c^\infty(\mathbb{R}^d)$ is contained in the principal ideal generated by u . Let $g \in C_c^\infty(\mathbb{R}^d)$, and let $K = \text{supp } g$. The set K is compact. Since u is continuous and strictly positive, there is $m_K > 0$ such that $u(x) \geq m_K$ for all $x \in K$. If $M_K = \|g\|_\infty$, then

$$|g(x)| \leq \frac{M_K}{m_K} u(x) \quad (x \in K),$$

and outside K the left hand side is zero. Thus

$$-\lambda u \leq g \leq \lambda u \quad \text{a.e. on } \mathbb{R}^d, \quad \lambda = \frac{M_K}{m_K}.$$

Hence $g \in X_u$.

The standard Sobolev density theorem gives

$$\overline{C_c^\infty(\mathbb{R}^d)}^{W^{k,p}} = W^{k,p}(\mathbb{R}^d) \quad (1 \leq p < \infty).$$

Since $C_c^\infty(\mathbb{R}^d) \subseteq X_u$, the principal ideal X_u is dense in $W^{k,p}(\mathbb{R}^d)$. By definition, u is a quasi-interior point. The implication from quasi-interior to almost interior is recorded in the source paper immediately after Definition 2.4. $\square$

Limitations

This resolves the Sobolev-space question in Example 2.20 of the source paper. It does not address the broader Open Problem 2.5, which asks whether every almost interior point must be quasi-interior in arbitrary ordered Banach spaces with generating cone, or under the additional assumption of normality.

Novelty and Verification Notes

The run's cheap indexes contain no prior packet for arXiv:1901.03306 or this Sobolev question. Local corpus searches for Sobolev quasi-interior phrases found the source question but no existing answer. Web searches for the exact Sobolev/quasi-interior phrases likewise found no later resolution.

The proof uses only the standard density of $C_c^\infty(\mathbb{R}^d)$ in $W^{k,p}(\mathbb{R}^d)$ for $1 \leq p < \infty$. No numerical computation is involved. PDF rendering tools capable of cropping the exact source page were not available in this environment; the source location is therefore given by the exact Example 2.20 passage and parsed-source lines 823–837.

References

- [1] R. A. Adams and J. J. F. Fournier, *Sobolev Spaces*, 2nd ed., Pure and Applied Mathematics 140, Academic Press, 2003.
- [2] J. Glück and M. R. Weber, *Almost Interior Points in Ordered Banach Spaces and the Long-Term Behaviour of Strongly Positive Operator Semigroups*, arXiv:1901.03306.